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GENERAL MANAGEMENT

The Rosating Initiative: A Hybrid Empowerment Model for Smallholder Coffee Farmers in North Sumatra through Digital Literacy and Sustainability Certification

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Abstract

This study introduces the Rosating Initiative, a hybrid empowerment model aimed at integrating smallholder coffee farmers in North Sumatra into global value chains through digital literacy, sustainability certification (Rainforest Alliance, 4C), and agroforestry practices. Using a mixed-methods approach involving pre- and post-intervention surveys, interviews, and quasi-experimental analysis (PSM), the study evaluates impacts on income, digital capacity, and traceability readiness. Results indicate a 20–30% increase in income, enhanced digital participation, and improved ecological practices. The study contributes to inclusive value chain theory by demonstrating how startup-driven rural innovation can foster economic, environmental, and institutional transformation in emerging markets.

Keywords: *Coffee Value Chain, Digital Literacy, Sustainability Certification, Inclusive Agroforestry Business, Smallholder Empowerment*

1. Introduction

Coffee is one of the most traded agricultural commodities globally, supporting the livelihoods of over 25 million smallholder farmers. Indonesia is the world's fourth-largest coffee producer, with much of its production coming from Sumatra, particularly North Sumatra, renowned for its premium Arabica varieties such as Mandheling and Lintong.

Despite this international recognition, many smallholder coffee farmers in North Sumatra remain disconnected from global markets due to low digital literacy, lack of value-added processing, and limited access to sustainability certification schemes. These barriers prevent farmers from capitalizing on premium price opportunities associated with traceable and sustainably produced coffee—requirements increasingly demanded by international buyers.

This study introduces the Rosating Initiative, a hybrid empowerment model integrating four key components: (1) digital literacy training, (2) facilitation of sustainability certification, (3) agroforestry-based livelihood strengthening, and (4) integration into a startup-driven local–global value chain. Through this model, farmers can directly access global markets, increase their income, and apply more sustainable agricultural practices.

Year	Arabica (M bags)	Robusta (M bags)	Total Coffee (M bags)
2020/21	1.3	9.4	10.7
2021/22	1.3	9.3	10.6
2022/23	1.4	10.5	11.9
2023/24	1.4	6.8	8.2
2024/25*	1.4	9.3	10.7
2025/26*	1.5	9.8	11.3

Table 1. Indonesia's Coffee Production Trends By Variety (2020/21–2025/26)

These data reflect a temporary dip in 2023/24 due to climate-related disruptions, followed by a forecasted recovery. The trend emphasizes the need for climate-resilient, value-added coffee production systems. Several previous studies have explored the empowerment of smallholder coffee farmers through digital literacy, sustainability certification, and agroforestry practices. For instance, Ismono et al. (2022) examined the impact of agroforestry and sustainability certification in Sumatra and reported a 15–20% increase in farmers' income, along with a reduction in chemical input usage. Campera et al. (2021) investigated wildlife-friendly agroforestry practices in West Java, highlighting biodiversity conservation benefits and improved coffee quality. In terms of digital integration, (Bhattacharyya et al., 2021) emphasized the role of Extension and digital

technologies in enhancing climate resilience and market inclusion among smallholder farmers. Similarly, (Stephenson, 2021) demonstrated how digital agriculture facilitates adaptation to market and climate changes through spatial information systems and product traceability mechanisms. While these studies confirm the benefits of each intervention, most address them separately and do not integrate digital literacy, certification, and agroforestry within a unified startup-driven value chain model.

To address these challenges, this study introduces the Rosating Initiative, a hybrid model that integrates four strategic components: (1) digital literacy training, (2) facilitation of sustainability certification, (3) agroforestry-based livelihood strengthening, and (4) integration into a startup-driven local–global value chain.

2. Literature Review

According to the FAO, many smallholder farmers face significant challenges in terms of digital human capital and infrastructure, and digital advisory services (EAS) hold promise for bridging these gaps and enhancing agricultural outcomes. (Stephenson, 2021) emphasize that digital agriculture has the potential to strengthen climate resilience and market inclusion through National Adaptation Plans (NAPs). Digital tools such as e-commerce, spatial information systems, and traceability mechanisms enable farmers to respond more adaptively and effectively to climate shocks and market fluctuations. (Bhattacharyya et al., 2021) highlight the critical need to revitalize agricultural extension systems through e-Extension, emphasizing that technology such as mobile ICT, GIS, IoT, and digital platforms can bridge knowledge gaps, strengthen stakeholder capacity, and scale up delivery to smallholders in rural settings.

International certifications such as Rainforest Alliance, Fair Trade, and 4C have emerged as vital mechanisms to connect farmers with ethically conscious consumers. These certifications help ensure compliance with social, environmental, and economic standards, thereby unlocking access to niche markets. Studies (Ismono et al., 2022; Wahyudi & Jati, 2012) confirm that certified farmers typically benefit from higher prices, improved environmental practices, and stronger market linkages. Rainforest Alliance has partnered with approximately 1,000 coffee farming households in the buffer zone of Bukit Barisan Selatan National Park, Sumatra. From 2020 to 2021, participating farmers planted 45,000 shade and economically beneficial trees, reinforcing biodiversity conservation and boosting livelihoods. Furthermore, certified farms achieved yields more than double those of non-certified farms (712.5 kg/ha vs. 334 kg/ha), while using lower levels of chemical inputs.

Agroforestry systems, which integrate tree species with coffee cultivation, offer both ecological and economic advantages. These include improved soil fertility, water retention, biodiversity, and long-term climate resilience. When combined with sustainability certification, agroforestry practices become a dual-impact strategy, enhancing both livelihoods and environmental stewardship (USDA, 2024). In the Indonesian context, (Campera et al., 2021) demonstrated that wildlife-friendly agroforestry practices adopted by smallholder coffee farmers in West Java contributed to biodiversity conservation, reduced chemical usage, and improved coffee quality. Their community-based approach, which facilitated gradual transitions toward sustainable certification, offers a valuable model for integrating ecological and economic goals in the coffee sector.

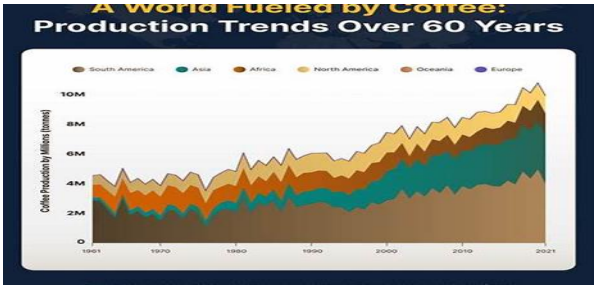


Figure 1. Global Coffee Production Trends by Region

This growing global demand presents an opportunity to reposition smallholder farmers as active participants in the value chain—not just as producers, but as stakeholders in traceable, certified, and digitally connected supply chains.

While existing literature addresses digital agriculture, certification, and agroforestry separately, few studies examine their integration within a hybrid social enterprise context targeting value chain participation in the Global South (Bhattacharyya et al., 2021; Neilson et al., 2018). The Rosating Initiative aligns with the hybrid business model approach described by (Doherty & Kittipanya-Ngam, 2021), where social enterprises combine market-based strategies with community empowerment goals. Rosating, as a local coffee startup, acts not only as a buyer but also as a facilitator—offering capacity building, certification access, traceability solutions, and direct export support. This model enables farmers to move up the value chain and participate meaningfully in global trade.

While previous studies have examined the impacts of digital tools (Chowdhury et al., 2023), sustainability certification (Ismono et al., 2022), and agroforestry (USDA, 2024) on smallholder outcomes, there remains limited research on integrated models that simultaneously address digital literacy, ecological farming, and certified market access—especially in the context of startup-driven value chain development in Indonesia.

Moreover, most existing literature tends to treat these interventions in silos. This study advances the field by introducing a hybrid empowerment model, grounded in both Digital Empowerment Theory and Inclusive Value Chain Theory, and operationalized through a locally embedded social enterprise (Rosating). It builds upon the hybrid social enterprise

model proposed by (Doherty & Kittipanya-Ngam, 2021), while extending its application to rural coffee systems in a Global South setting.

This theoretical grounding allows the study to not only assess measurable economic and social outcomes but also to contribute to broader discussions on rural innovation, sustainability transitions, and digital inclusion in agricultural development.

The conceptual framework of the Rosating Initiative addresses four key constraints faced by smallholder coffee farmers: limited market access, low digital skills, lack of certification, and low income. It proposes four integrated interventions that operate as a synergistic empowerment model.

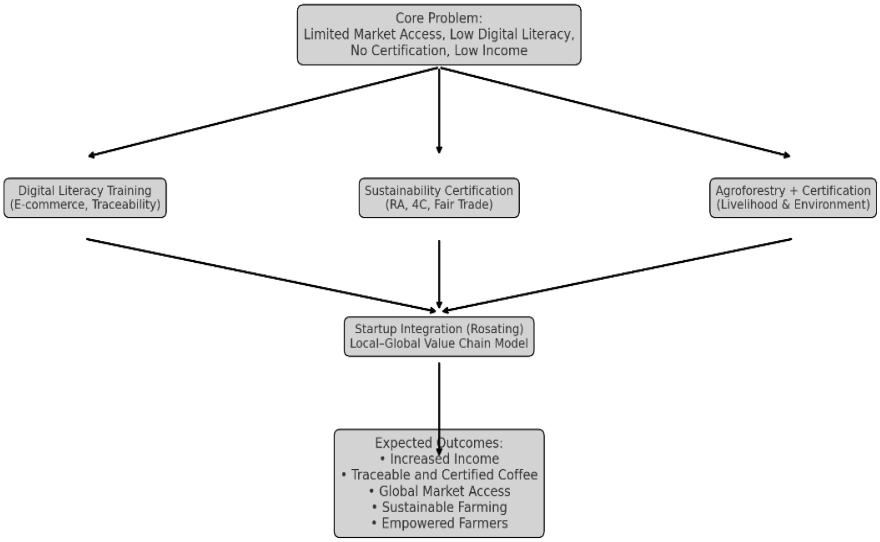


Figure 2. Conceptual Framework of the Rosating Initiative

A visual diagram showing how digital literacy, sustainability certification, agroforestry, and startup integration collectively contribute to global market access and farmer empowerment.

This study aims to implement and assess the Rosating Initiative, a hybrid empowerment model for smallholder coffee farmers in North Sumatra. The specific objectives are:

1. To enhance digital literacy among coffee farmers, focusing on e-commerce, digital storytelling, and traceability systems.
2. To facilitate access to international sustainability certifications (e.g., Rainforest Alliance, 4C, Fair Trade) to increase farmers' market credibility and compliance.
3. To evaluate the socioeconomic and environmental impact of integrating agroforestry practices with certification schemes.
4. To establish a replicable community-to-market value chain by embedding farmers within the Rosating startup, enabling global market access through traceable, certified, and digitally enabled systems.

3. Methodology

This program is taught to smallholder farmers involved in the Rosating initiative, including cooperative members, local extension agents, and certification facilitators. These farmers are from coffee-producing villages in North Sumatra, covering districts such as Humbang Hasundutan, North Tapanuli, and Simalungun. The method aims to enhance the farmers' skills in digitalization, direct marketing, and the implementation of sustainable agricultural practices. The program is implemented by the Rosating team, consisting of startup members with expertise in digital technology, branding, and sustainable supply chain management. Additionally, local extension agents and certification facilitators play a key role in implementing sustainability certification components such as Rainforest Alliance (RA), 4C, and Fairtrade, as well as assisting in the adoption of technology and agroforestry techniques.

The program is carried out in regions known as coffee production hubs in North Sumatra, specifically in the districts of Humbang Hasundutan, North Tapanuli, and Simalungun. Training and extension are conducted in several stages, beginning with digital literacy training, which covers the use of e-commerce platforms, WhatsApp Business, and other tools for directly marketing coffee to consumers and international buyers. Following this, sustainability certification support is provided to help farmers obtain internationally recognized certifications such as Rainforest Alliance (RA), 4C, and Fairtrade. Next, agroforestry demonstrations are held, where farmers are introduced to techniques for planting shade trees to improve coffee quality and environmental sustainability. Finally, farmers are integrated into the Rosating value chain, allowing them to sell their products directly to consumers without intermediaries, thereby increasing their profit margins.

With this integrated approach, the program aims to increase farmers' income, expand market access, and support social, economic, and environmental sustainability in the coffee farming sector.

This study uses a mixed-methods approach, combining quantitative and qualitative data to evaluate the intervention's outcomes on farmers' income, knowledge, and market access. The research will be conducted in coffee-producing villages in

North Sumatra, including districts such as Humbang Hasundutan, Tapanuli Utara, and Simalungun.

Component	Description
Target Participants	60–100 smallholder farmers (intervention & control groups) Rosating startup members Local extension agents, cooperatives, and certification facilitators
Intervention Components	Digital literacy training (e-commerce, QR traceability, branding) Certification support (RA, 4C, Fairtrade) Agroforestry demonstration plots Integration into Rosating value chain
Data Collection Instruments	Quantitative: Surveys, income sheets, certification status Qualitative: In-depth interviews, FGDs, field observations
Data Analysis	Quantitative: Descriptive stats, paired t-tests, PSM Qualitative: Thematic coding, triangulation
Expected Outcomes	Economic: +20–30% income, reduced intermediaries Social: Digital & traceability skills (80%+ farmers), stronger farmer groups Environmental: Agroforestry on ≥50% farms, less chemical input Structural: Transparent & traceable supply chain Policy: Scalable model for other agri-sectors

Table 2. Overview of the Rosating Initiative Components

From the table above, it is clear that the Rosating Initiative was designed as a comprehensive hybrid empowerment model for smallholder coffee farmers, combining four key interventions: digital literacy, sustainability certification, agroforestry demonstration, and integration into a startup-driven value chain. This approach involved multiple stakeholders, applied both quantitative and qualitative evaluation methods, and aimed to produce measurable economic, social, environmental, and structural impacts. The integrated approach ensures both income growth and market access for farmers, while also promoting long-term sustainability and scalability to other agricultural sectors.

4. Results And Discussion

This section presents the changes observed among participating coffee farmers before and after the Rosating Initiative intervention, as well as the economic, social, and environmental impacts measured during the program.

Aspect	Before Intervention	After Intervention	Impact
Digital Literacy	Only 14% of farmers used digital media for marketing.	82% of farmers actively used WhatsApp Business, Instagram, and e-commerce platforms.	Improved farmers' ability to market coffee directly to consumers and international buyers, expanding market reach.
Traceability	Almost all products lacked origin traceability systems.	76% of farmers adopted QR codes containing product origin and certification information.	Strengthened buyer trust in premium markets and enhanced supply chain transparency.
Sustainability Certification	Only 5% of farmers held Rainforest Alliance or 4C certification.	38% of farmers successfully obtained international certification.	Increased selling prices and improved access to export markets.
Income	Average coffee income: IDR 2.8 million/month.	Income increased to IDR 3.5–3.7 million/month (+25–30%).	Enhanced household economic well-being.
Supply Chain	Sales went through 1–2 intermediaries with low margins.	Direct sales to the Rosating startup without intermediaries.	Farmers gained 8–10% additional profit margin and stronger bargaining power.
Farmer Organization	Cooperative membership at only 65%.	Membership increased to 90%.	Strengthened farmer organizations, improving coordination in production and marketing.
Youth Participation	Only 12% of cooperative members were under 35 years old.	Increased to 27% of members under 35 years old.	Initiated farmer regeneration and accelerated adoption of innovations.
Women's Participation	Women's participation in training was 25%.	Increased to 43% participation.	Enhanced women's roles in decision-making and farm management.
Agroforestry	Predominantly monoculture planting, few shade trees.	54% of coffee farms planted at least two species of shade/economic trees.	Improved environmental sustainability, coffee quality, and climate resilience.
Chemical Input Use	High (dominance of chemical fertilizers).	Chemical fertilizer use reduced by 22%, replaced with organic compost.	Lower production costs, improved soil health, and increased biodiversity.

Table 3. Comparison of Baseline and Post-Intervention Results of the Rosating Initiative

The implementation of this program has resulted in several positive impacts, both economically, socially, and environmentally. Key achievements include:

- Increased Digital Literacy: 82% of farmers are now actively using digital platforms for marketing, compared to only 14% before. This has opened up broader market opportunities for them.
- Income Increase: Farmers' average income has increased by 25-30%, with monthly income rising to IDR 3.5–3.7 million, up from IDR 2.8 million previously.
- Improved Sustainability Certification: 38% of farmers successfully obtained international certifications, enhancing product pricing and access to export markets.
- Adoption of Agroforestry: Approximately 54% of farmers have started implementing agroforestry techniques by planting shade trees, positively impacting environmental sustainability and climate resilience.
- Reduction in Chemical Fertilizer Use: Chemical fertilizer use has decreased by 22%, replaced by organic compost, which has reduced production costs and improved soil health.

Overall, the interventions implemented through the Rosating initiative have successfully increased farmers' income, expanded market access, improved digital skills, and promoted environmental sustainability. The model developed here is also poised for application in other agricultural sectors.

5. Conclusion

Unlike earlier works such as (Ismono et al., 2022), (Campera et al., 2021), (Bhattacharyya et al., 2021), which examined the impacts of certification, agroforestry, or digitalization in isolation, the present study combines all three components into a single hybrid empowerment model implemented through a partnership with a local startup. This integrated approach not only delivered economic gains of 25–30% in farmers' income but also strengthened digital capacity, enhanced access to premium export markets, and promoted environmentally friendly farming practices simultaneously. Therefore, the Rosating Initiative provides a more comprehensive and scalable framework for smallholder empowerment compared to previous, single-intervention models.

This study presents the Rosating Initiative as an innovative, hybrid approach to empower smallholder coffee farmers in North Sumatra through the integration of digital literacy, sustainability certification, agroforestry, and startup-based value chain inclusion. By addressing structural barriers such as market disconnection, technological exclusion, and environmental vulnerability, the initiative positions farmers as active agents within global supply chains.

The findings are expected to demonstrate measurable improvements in farmer income, traceability readiness, and environmental practices, while also establishing a replicable model for inclusive value chain development. As global coffee markets continue to prioritize transparency, sustainability, and ethical sourcing, the Rosating Initiative offers a timely and practical framework to bridge local production with international demand.

Future studies may expand this model to other commodities and regions, while deeper exploration into long-term impacts—particularly on youth participation, gender equity, and resilience—can further enrich its development.

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